Iteration L3: Moving to Production

Running in production is so much more than just deploying your application. It's about keeping your application running smoothly and keeping your data safe. If your server goes down, a new one can be started in minutes. If your data is lost, it's lost forever.

Backing Up Your Database

If you look at your `config/database.yml`, you'll see that it contains definitions for four databases: primary, cache, queue, and cable. These are stored in a storage directory, which is mapped to a directory on your host machine. This is where your data is stored. It's quite possible to back up all four, but for now we're going to focus on the primary database. The primary database is the one that contains your users, products, and orders. It's the one that you can't afford to lose. Because you set up a volume, your data is safe as long as your host machine is running. But what if your host machine goes down?

Docker stores its data in a directory called `/var/lib/docker`. Inside that directory is a directory called `volumes`. You created a volume called `work_storage`. Inside the `volumes` directory is a file named `production.sqlite3`. This is your database.

Litestream[122] is a tool that can be used to back up your database. It's free and open source. It's easy to set up and can be used to back up to the S3 Object Storage you've already set up. It can also be used to restore your database to a point in time.

We could run it in your Kamal container or as a Kamal accessory, but instead, we're going to run it on your host machine. This ensures it's always running and we don't have to worry about multiple instances of it running as you deploy new versions of your application.